

Marbles on a tree

Tags:

Time Limit: 1 s Memory Limit: 128 MB
 Submission: 16 AC: 9 Score: 99.62

Submit

Codes

AI Code Tips

Description

n
 boxes are placed on the vertices of a rooted tree, which are numbered from 1 to n .
 n
 $, 1 \leq v \leq n$.
 n
 ≤ 10000 . Each box is either empty or contains a number of marbles; the total number of marbles is m .
 n
 .



The task is to move the marbles such that each box contains exactly one marble. This is to be accomplished by a sequence of moves; each move consists of moving one marble to a box at an adjacent vertex. What is the minimum number of moves required to achieve the goal?

The input contains a number of cases. Each case starts with the number n followed by n lines. Each line contains at least three numbers which are: v the number of a vertex, followed by the number of marbles originally placed at vertex v followed by a number d which is the number of children of v , followed by d numbers giving the identities of the children of v .

The input is terminated by a case where $n = 0$ and this case should not be processed.

For each case in the input, output the smallest number of moves of marbles resulting in one marble at each vertex of the tree.

Samples

input	Copy
<pre> 9 1 2 3 2 3 4 2 1 0 3 0 2 5 6 4 1 3 7 8 9 5 3 0 </pre>	

```
6 0 0
7 0 0
8 2 0
9 0 0
9
1 0 3 2 3 4
2 0 0
3 0 2 5 6
4 9 3 7 8 9
5 0 0
6 0 0
7 0 0
8 0 0
9 0 0
9
1 0 3 2 3 4
2 9 0
3 0 2 5 6
4 0 3 7 8 9
5 0 0
6 0 0
7 0 0
8 0 0
9 0 0
0
```

output

Copy

```
7
14
20
```

Submit

Codes

[HZNUOJ](#) is based on [HUSTOJ](#)

[--- Maintainer List ---](#)

Server Time: 2025-8-26 15:07:00